

Identifying-the-best-recruiting-source.R

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```
# Load the packages
library("readr")
library("dplyr")

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library("ggplot2")

# Import the recruitment data
recruitment <- read_csv("~/Desktop/Data/recruitment_data.csv" )

## Rows: 446 Columns: 4

## -- Column specification -----
## Delimiter: ","
## chr (1): recruiting_source
## dbl (3): attrition, performance_rating, sales_quota_pct
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

head(recruitment)

## # A tibble: 6 x 4
##   attrition performance_rating sales_quota_pct recruiting_source
##   <dbl>         <dbl>         <dbl> <chr>
## 1         1             3             1.09 Applied Online
## 2         0             3             2.39 <NA>
## 3         1             2             0.498 Campus
## 4         0             2             2.51 <NA>
## 5         0             3             1.42 Applied Online
## 6         1             3             0.548 Referral
```

```
# Get an overview of the recruitment data
summary(recruitment)
```

```
##      attrition      performance_rating sales_quota_pct      recruiting_source
## Min.       :0.000    Min.       :1.000      Min.       :-0.7108    Length:446
## 1st Qu.:0.000    1st Qu.:2.000      1st Qu.: 0.5844    Class :character
## Median :0.000    Median :3.000      Median : 1.0701    Mode  :character
## Mean      :0.213    Mean      :2.895      Mean       : 1.0826
## 3rd Qu.:0.000    3rd Qu.:3.000      3rd Qu.: 1.5325
## Max.       :1.000    Max.       :5.000      Max.        : 3.6667
```

```
# See which recruiting sources the company has been using
recruitment %>% count(recruiting_source)
```

```
## # A tibble: 5 x 2
##   recruiting_source      n
##   <chr>              <int>
## 1 Applied Online      130
## 2 Campus              56
## 3 Referral           45
## 4 Search Firm         10
## 5 <NA>               205
```

```
# Find the average sales quota attainment for each recruiting source
avg_sales <- recruitment %>%
  group_by(recruiting_source)%>%
  summarize(avg_sales_quota_pct = mean(sales_quota_pct))
```

```
# Display the result
avg_sales
```

```
## # A tibble: 5 x 2
##   recruiting_source avg_sales_quota_pct
##   <chr>              <dbl>
## 1 Applied Online      1.06
## 2 Campus              0.908
## 3 Referral           1.02
## 4 Search Firm         0.887
## 5 <NA>               1.17
```

```
# Find the average attrition for the sales team, by recruiting source, sorted from lowest attrition rate
avg_attrition <- recruitment %>%
  group_by(recruiting_source)%>%
  summarize(attrition_rate = mean(attrition))%>%
  arrange(attrition_rate)
```

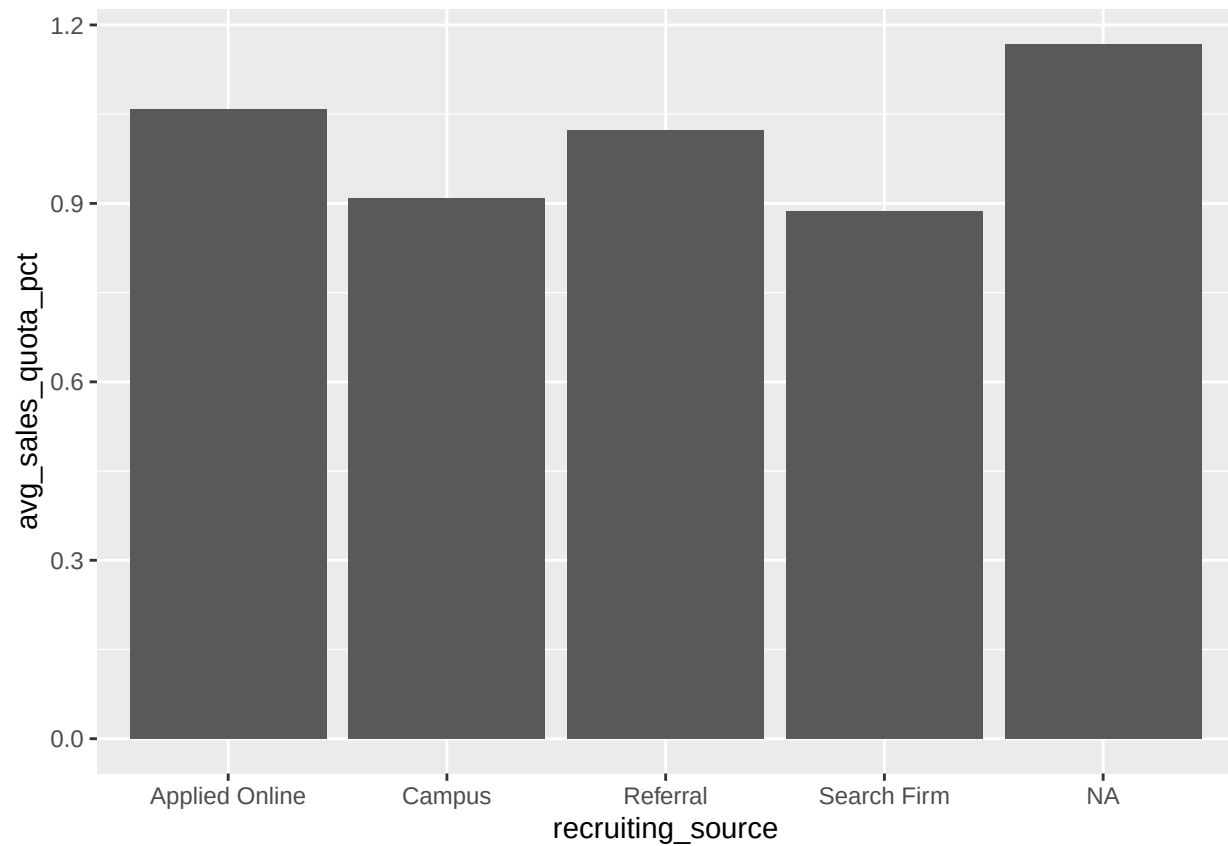
```
# Display the result
avg_attrition
```

```
## # A tibble: 5 x 2
```

```
## recruiting_source attrition_rate
## <chr> <dbl>
## 1 <NA> 0.132
## 2 Applied Online 0.246
## 3 Campus 0.286
## 4 Referral 0.333
## 5 Search Firm 0.5
```

```
# Plot the bar chart
```

```
ggplot(avg_sales, aes(x=recruiting_source, y= avg_sales_quota_pct)) +
  geom_col()
```



```
# Plot the bar chart
```

```
ggplot(avg_attrition, aes(x=recruiting_source, y= attrition_rate)) +
  geom_col()
```

